MATH1014 Calculus II Tutorial
Week 2

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1 Review

1.1 Fundamental Theorem of Calculus (FTC)

1. If $F'(x) = f(x)$ on an interval, then
$$\int_a^b f(t)\,dt = F(b) - F(a).$$
2. If f is continuous and $A(x) = \int_a^x f(t)\,dt$, then
$$A'(x) = f(x).$$
3. Indefinite integrals: $\displaystyle\int f(x)\,dx = F(x) + C$ where $F'(x) = f(x)$.

1.2 Area Between Curves

1. Using dx (vertical slices): if $y = f(x)$ is above $y = g(x)$ on $[a, b]$, then
$$\text{Area} = \int_a^b \big(f(x) - g(x)\big)\,dx.$$
2. Using dy (horizontal slices): if $x = h(y)$ is right of $x = k(y)$ on $[c, d]$, then
$$\text{Area} = \int_c^d \big(h(y) - k(y)\big)\,dy.$$
3. If the top/bottom (or right/left) changes, split the interval at intersection points.

1.3 Volumes

1. Slicing (cross-sectional area): if the slice area is $A(x)$ for $x \in [a, b]$, then
$$\text{Vol} = \int_a^b A(x)\,dx.$$
2. Cylindrical shells: rotating a vertical strip about the y-axis,
$$\text{Vol} = \int_a^b 2\pi\, r(x)\, h(x)\,dx.$$

2 Exercises

Exercise 2.1 (Warm-up: identifying "top" and "bottom"). Evaluate the area of the following regions enclosed by the given curves.

1. $y = \frac{1}{x}$, $y = x$, $x = 2$.
2. $y^2 = 4(x+1)$, $y^2 = 4(1-x)$.
3. $y = \cos x$, $y = \sin 2x$, $x = 0$, $x = \frac{\pi}{2}$.

Solution. **Idea.** For area between curves using dx, we aim for

$$\text{Area} = \int_{x=\text{left}}^{x=\text{right}} \big(y_{\text{top}}(x) - y_{\text{bottom}}(x)\big)\, dx,$$

and we split the interval whenever the identity of the top/bottom curve changes.

(1) The curves $y = x$ and $y = \frac{1}{x}$ intersect at $x = 1$ (with $y = 1$). On $[1, 2]$, the line is above the hyperbola, so

$$\text{Area} = \int_1^2 \left(x - \frac{1}{x}\right) dx = \left(\frac{x^2}{2} - \ln x\right)\Big|_1^2 = \frac{3}{2} - \ln 2.$$

(2) Rewrite each parabola as x in terms of y:

$$y^2 = 4(x+1) \Rightarrow x = \frac{y^2}{4} - 1, \qquad y^2 = 4(1-x) \Rightarrow x = 1 - \frac{y^2}{4}.$$

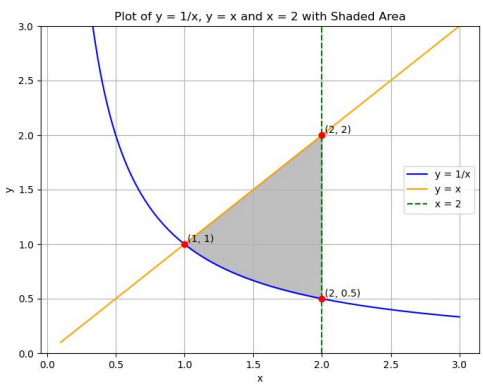
It is symmetric about the x-axis, so compute the upper half ($y \geq 0$) and double:

$$\text{Area} = 2\int_0^2 \left[\left(1 - \frac{y^2}{4}\right) - \left(\frac{y^2}{4} - 1\right)\right] dy = 2\int_0^2 \left(2 - \frac{y^2}{2}\right) dy = \frac{16}{3}.$$

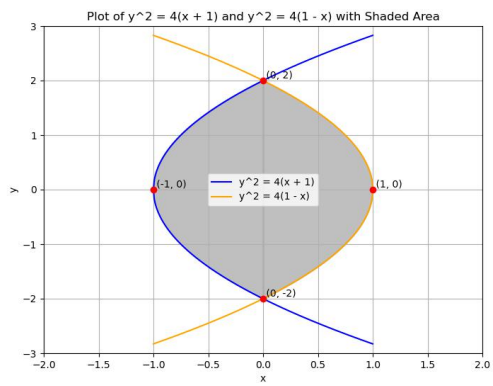
(3) First find intersection points. Solve $\cos x = \sin 2x = 2\sin x \cos x$. Either $\cos x = 0$ giving $x = \frac{\pi}{2}$, or $1 = 2\sin x$ giving $x = \frac{\pi}{6}$ (within $[0, \frac{\pi}{2}]$). On $[0, \frac{\pi}{6}]$, $\cos x \geq \sin 2x$; on $[\frac{\pi}{6}, \frac{\pi}{2}]$, $\sin 2x \geq \cos x$. Hence

$$\text{Area} = \int_0^{\frac{\pi}{6}} (\cos x - \sin 2x)\, dx + \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (\sin 2x - \cos x)\, dx = \frac{1}{2}.$$

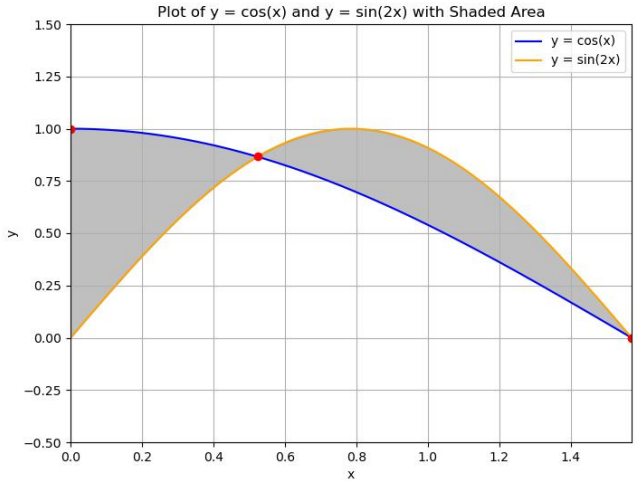
Checkpoint. An equivalent one-line setup is $\text{Area} = \int_0^{\frac{\pi}{2}} |\cos x - \sin 2x|\, dx$.



Sketch for (1).



Sketch for (2).



Sketch for (3).

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Exercise 2.2 (Area with parameters / splitting regions)**.** Consider the region enclosed by $y = \frac{1}{x^2}$, $y = -\frac{4}{x^3}$, $x = 1$, $x = 4$.

1. Evaluate the area of the region.
2. Find a line $x = a$ such that the area of the left part divided by this line equals 2.
3. Find a line $y = b$ such that the ratio of area for the upper part and that for the lower part is $1 : 6$.

Solution. **Idea.** On $[1, 4]$, we have $\frac{1}{x^2} > 0$ and $-\frac{4}{x^3} < 0$, so the total area is the integral of (top minus bottom).

(1)

$$\text{Area} = \int_1^4 \left(\frac{1}{x^2} - \left(-\frac{4}{x^3}\right)\right) dx = \int_1^4 \left(\frac{1}{x^2} + \frac{4}{x^3}\right) dx = -\frac{1}{x} - \frac{2}{x^2}\bigg|_1^4 = \frac{21}{8}.$$

(2) Let $A(a) = \int_1^a \left(\frac{1}{x^2} + \frac{4}{x^3}\right) dx$. Solve $A(a) = 2$:

$$-\frac{1}{x} - \frac{2}{x^2}\bigg|_1^a = 2 \;\Rightarrow\; -\frac{1}{a} - \frac{2}{a^2} + 3 = 2 \;\Rightarrow\; a^2 - a - 2 = 0.$$

So $a = 2$ or $a = -1$; the interval condition $1 < a < 4$ forces $a = 2$.

(3) Split into the part above the x-axis and below the x-axis. Upper area:

$$A_{\text{upper}} = \int_1^4 \frac{1}{x^2} dx = -\frac{1}{x}\bigg|_1^4 = \frac{3}{4}.$$

Lower area:

$$A_{\text{lower}} = \text{Area} - A_{\text{upper}} = \frac{21}{8} - \frac{3}{4} = \frac{15}{8}.$$

The line $y = b$ splits the whole region into an *upper part* (above $y = b$) and a *lower part* (below $y = b$). The required ratio is

$$\text{Area}_{\text{above } y=b} : \text{Area}_{\text{below } y=b} = 1 : 6,$$

so the upper part has area $\text{Area}_{\text{above } y=b} = \frac{1}{7}\,\text{Area} = \frac{1}{7}\cdot\frac{21}{8} = \frac{3}{8}$.

Because $\frac{3}{8} < A_{\text{upper}} = \frac{3}{4}$, the cutting line must lie above the x-axis, so $0 < b < 1$. Express the upper curve as $x = \frac{1}{\sqrt{y}}$ (from $y = \frac{1}{x^2}$) and the right boundary as $x = 1$ at the left end of the region in xy-plane for $y \in [b, 1]$. Hence the area above $y = b$ in the upper region is

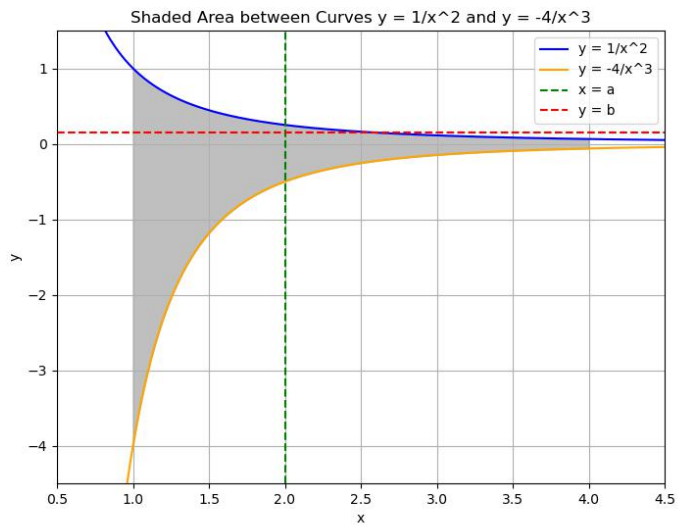
$$\int_b^1 \left(\frac{1}{\sqrt{y}} - 1\right) dy = (2\sqrt{y} - y)|_b^1 = 1 - 2\sqrt{b} + b.$$

Set it equal to $\frac{3}{8}$:

$$1 - 2\sqrt{b} + b = \frac{3}{8} \;\Rightarrow\; (\sqrt{b} - 1)^2 = \frac{3}{8} \;\Rightarrow\; \sqrt{b} = 1 - \frac{\sqrt{6}}{4},$$

so

$$b = \left(1 - \frac{\sqrt{6}}{4}\right)^2 = \frac{11 - 4\sqrt{6}}{8}.$$



Sketch for Exercise 2.

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Exercise 2.3 (Slicing method)**.** Evaluate the following volumes.

1. An elliptical frustum: the upper base is an ellipse with semi-axes a, b. The lower base is an ellipse with semi-axes A, B. The height is h.
2. An ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \le 1$.

Solution. **Idea.** For solids with "nice" cross-sections, write volume as

$$\text{Vol} = \int (\text{area of slice})\, d(\text{height}).$$

(1) Slice the frustum by planes parallel to the bases. At height x from the *top* base (so $x = 0$ at the top, $x = h$ at the bottom), the semi-axes change linearly:

$$a(x) = a + \frac{A - a}{h}x, \qquad b(x) = b + \frac{B - b}{h}x.$$

So the slice area is $\pi a(x)b(x)$ and

$$\begin{aligned}\text{Vol} &= \int_0^h \pi\left(a + \frac{A-a}{h}x\right)\left(b + \frac{B-b}{h}x\right)\,dx \\ &= \frac{\pi h}{6}\left(2AB + 2ab + Ab + aB\right).\end{aligned}$$

(2) For a fixed z (with $-c \le z \le c$), the cross-section is the ellipse

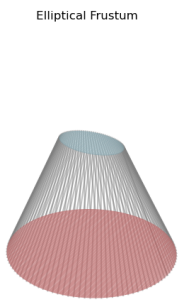
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 - \frac{z^2}{c^2},$$

whose semi-axes are $a\sqrt{1-\frac{z^2}{c^2}}$ and $b\sqrt{1-\frac{z^2}{c^2}}$. Therefore the slice area is

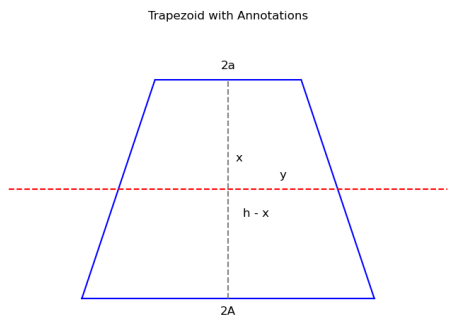
$$\pi ab\left(1 - \frac{z^2}{c^2}\right),$$

and

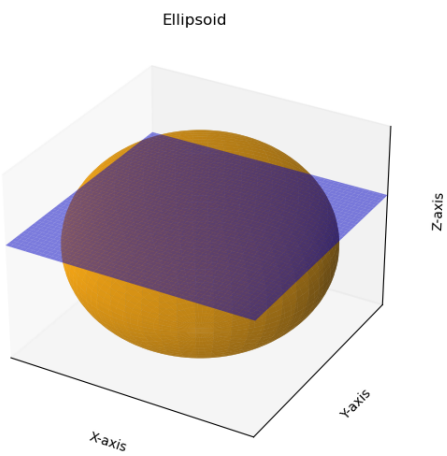
$$\text{Vol} = \int_{-c}^{c} \pi ab\left(1 - \frac{z^2}{c^2}\right)dz = \frac{4}{3}\pi abc.$$



Elliptical frustum.



Vertical section view.



Ellipsoid slicing.

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Exercise 2.4 (Washer method and symmetry)**.** Find the relationship between a and c such that the volumes generated by rotating the arc segment bounded by $y = x(x-a)$ and $y = 0$ around the x-axis within the intervals $[0, a]$ and $[a, c]$ are equal.

Solution. **Idea.** Rotating $y = f(x)$ around the x-axis gives a volume element $\pi(f(x))^2\,dx$ (disk method). Because the radius is squared, the sign of $f(x)$ does not matter.

On $[0, a]$, the function $x(x-a) \le 0$; on $[a, c]$, $x(x-a) \ge 0$. In both cases, the radius squared is $[x(x-a)]^2$.

So the volumes are

$$V_1 = \int_0^a \pi[x(x-a)]^2\,dx, \qquad V_2 = \int_a^c \pi[x(x-a)]^2\,dx.$$

Compute an antiderivative:

$$\int \pi[x(x-a)]^2\,dx = \pi\left(\frac{x^5}{5} - \frac{ax^4}{2} + \frac{a^2x^3}{3}\right).$$

Thus

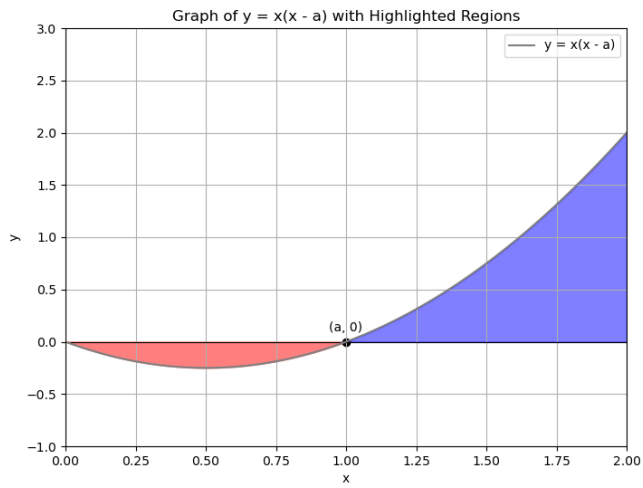
$$V_1 = \pi\left(\frac{a^5}{5} - \frac{a \cdot a^4}{2} + \frac{a^2 \cdot a^3}{3}\right) = \pi \cdot \frac{a^5}{30},$$

and

$$V_2 = \pi\left(\frac{c^5}{5} - \frac{ac^4}{2} + \frac{a^2c^3}{3}\right) - \pi \cdot \frac{a^5}{30}.$$

Setting $V_1 = V_2$ and cancelling π gives the relationship

$$2a^5 + 15ac^4 - 10a^2c^3 - 6c^5 = 0.$$



Sketch for Exercise 4.

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